EVALUATION OF NLP SYSTEMS

Lecture 4 - Annotation

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Illustration by Freepik Stories

- annotation in NLP

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TEST SET

a dataset for evaluation that should be

- *representative*
- *comprehensive*
- *consistently annotated*
- *unbiased (is it possible?)*
- *large enough (statistical purposes)*
- *sampled from the same probability distribution of the training data*

GOLD STANDARD

ground "truth" annotation in a data set

(hard to talk about truth when even humans disagree on correct labels)

Illustration by undraw

Annotation in NLP

metadata added to texts by humans, enriching it with linguistic information for machines to process (and understand?) language*

**can also be done automatically in some cases, but we'll focus on human annotation*

Illustration by undraw

some levels of linguistic annotation

discourse	cohesion in a text or dialogue
pragmatics	functions of utterances
semantics	meaning of words and sentences
syntax	word order and sentence structure
morphology	word formation and inflections
ortography	spelling (written language)
phonology	sounds (spoken language)

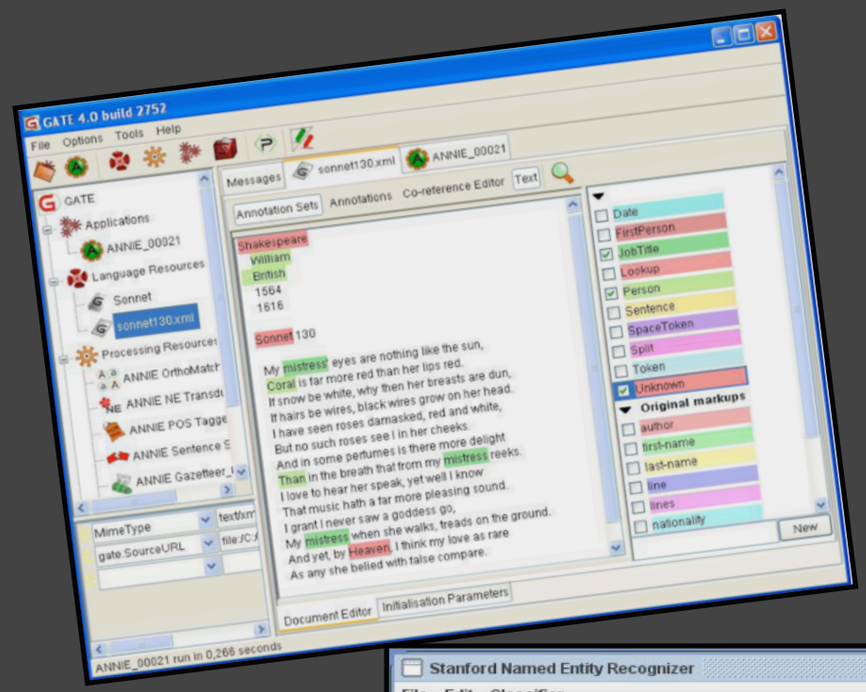
Wilcock (2009)

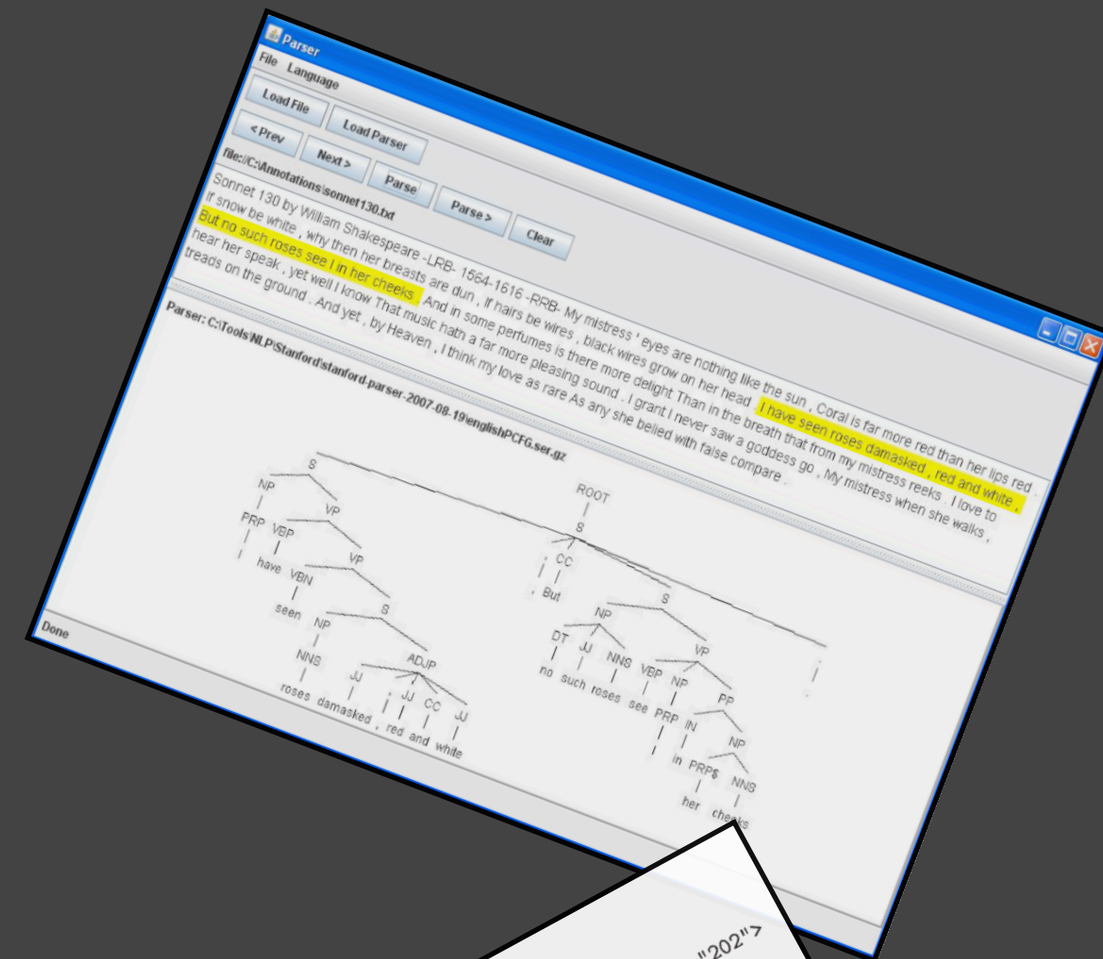
some levels of linguistic description

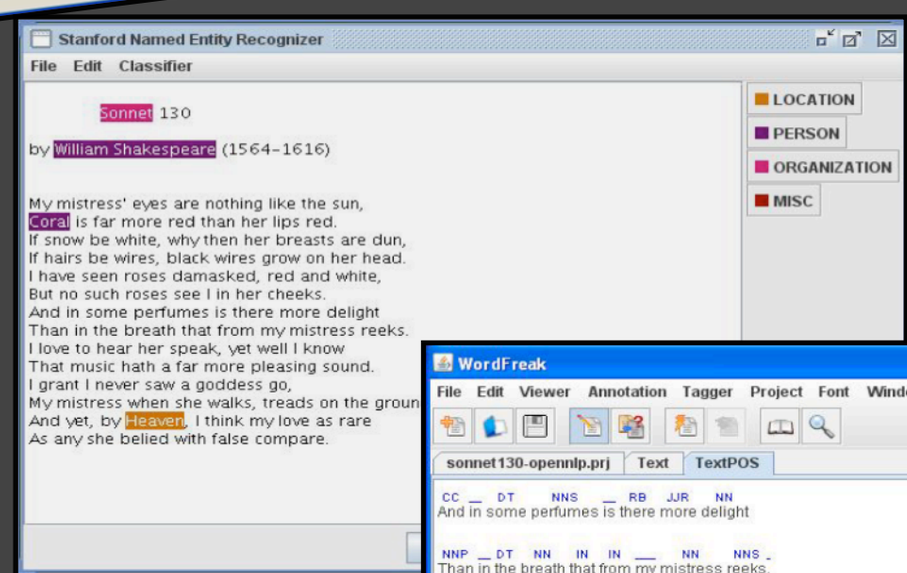
coreference resolution	linking references to same entities in a text
named entity recognition	identifying and labeling named entities
semantic analysis	labeling predicate-argument relations
syntactic parsing	analyzing constituent phrases in a sentence
part-of-speech tagging	labeling words with word categories
tokenization	segmenting text into words
sentence boundaries	segmenting text into sentences

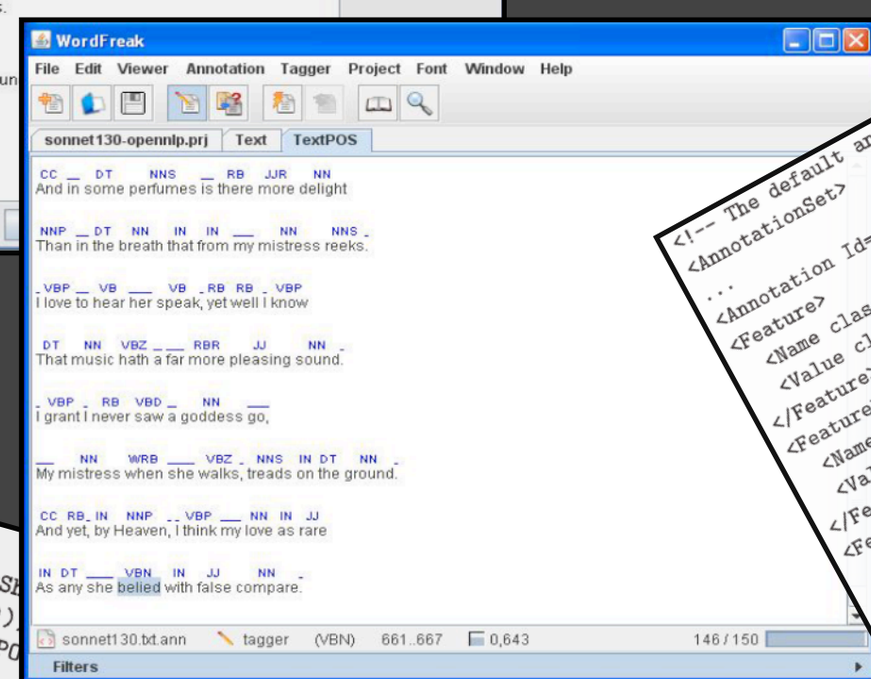
Wilcock (2009)

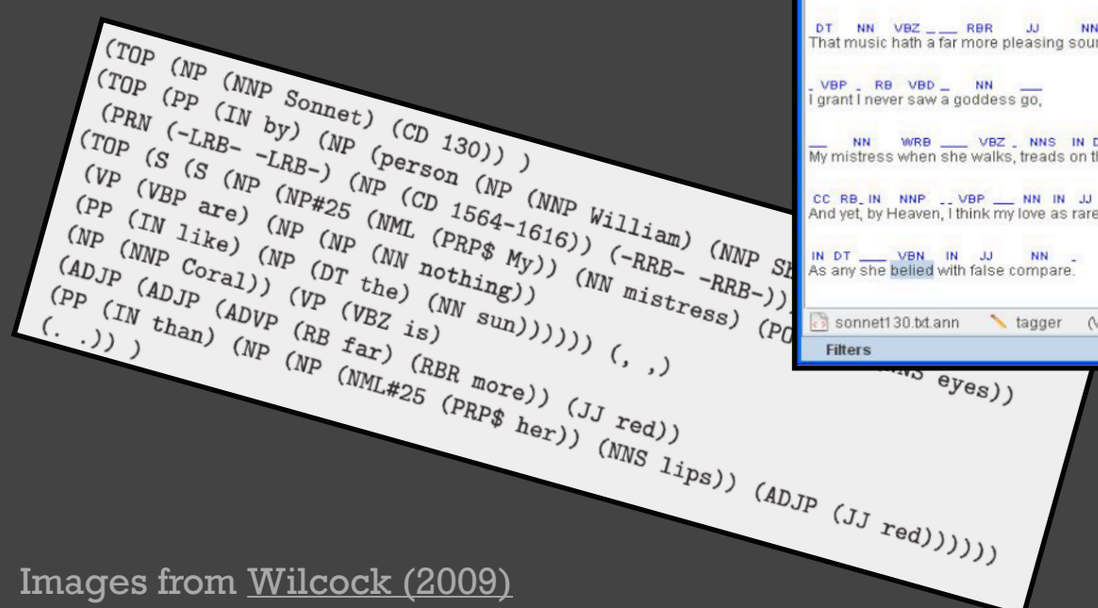
Examples

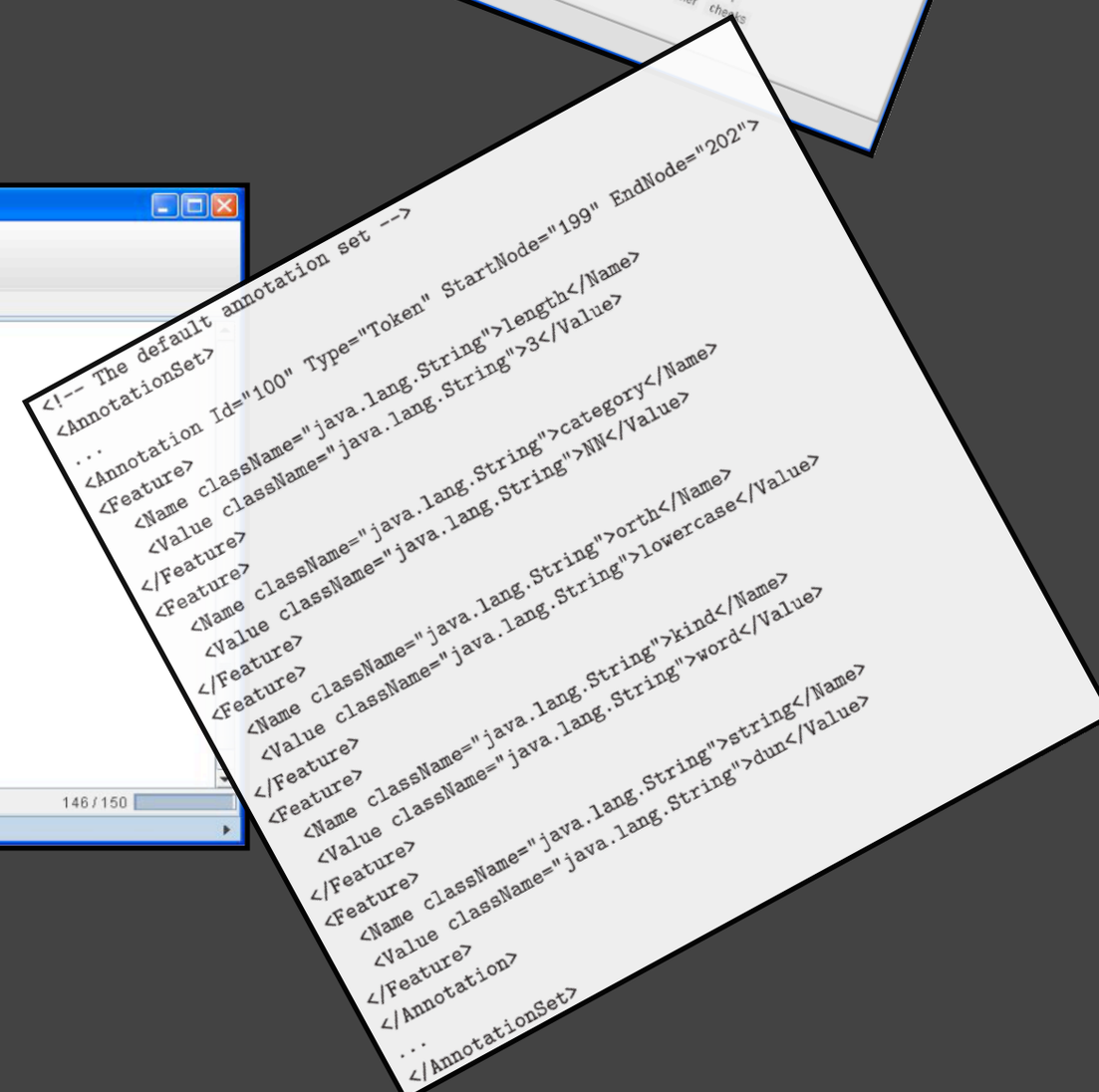












Images from Wilcock (2009)

evaluation usually depends on annotated datasets

where do they come from?

experts, researchers, annotators, students, system developers

the effort of many humans

it takes time, costs money and involves a lot of decisions

create & revise guidelines, train annotators, adapt tools, control quality, document

just what we need for automatic evaluation!

could we expect a system to perform better than humans annotators agreement?

but once annotated, we have gold standards and upper bounds

THE 7 QUESTIONS OF ANNOTATION

Hovy (2010)

- selecting a corpus
- instantiating the theory
- designing the interface
- selecting and training the annotators
- designing and managing the annotation procedure
- validating results
- delivering and maintaining the product

Illustration by Freepik Stories

THE ~~7~~ many QUESTIONS OF ANNOTATION

Hovy (2010)

- selecting a corpus
- instantiating the theory
- designing the interface
- selecting and training the annotators
- designing and managing the annotation procedure
- validating results
- delivering and maintaining the product

how does one obtain a balanced corpus to annotate, and when is a corpus balanced (and representative)?

how does one decide what specific phenomena to annotate?

how does one adequately capture the theory behind the phenomenon/a and express it in simple annotation instructions?

when hiring annotators, what characteristics are important?

how does one ensure that they are adequately (but not over- or under-) trained?

what interface(s) are best for each type of problem, and what should one know to avoid?

how can one ensure that the interfaces do not influence the annotation results?

how does one evaluate the results?

at which cutoff points should one redesign or re-do the annotations?

what are the appropriate agreement measures?

how does one establish a simple, fast, and trustworthy annotation procedure?

how and where can active learning help?

how and when does one apply measures to ensure that the procedure remains on track?

how should one report the annotation effort and results for best impact?

when, and to whom, should one release the corpus?

how should one formulate and store the results?

all these questions demand decisions and choices

and how it influences the system's performace

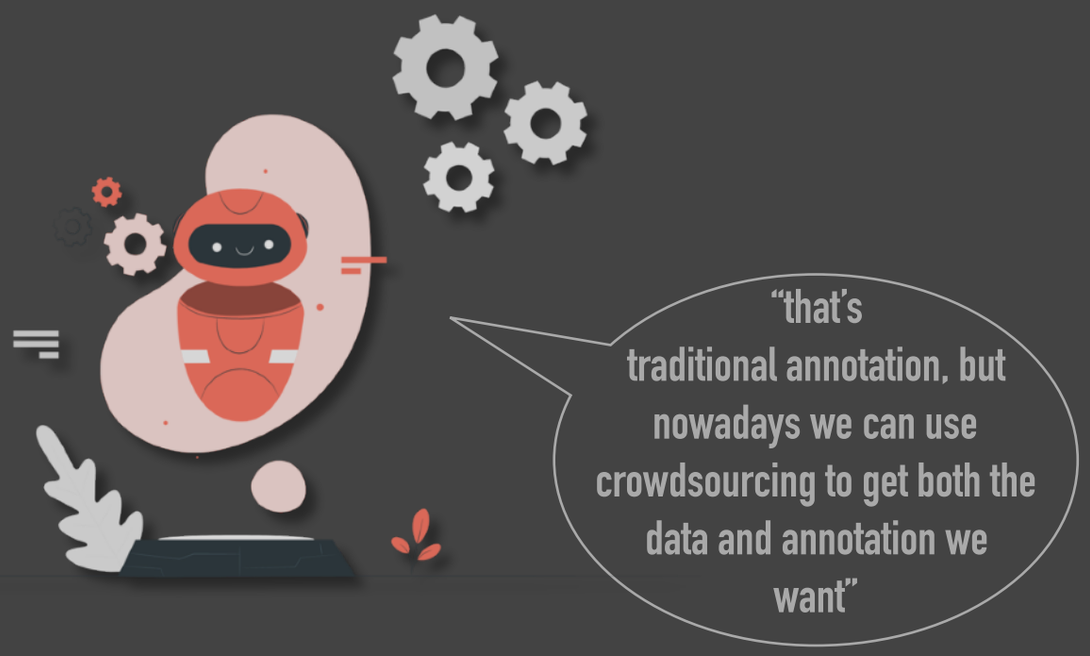
each may have an impact on NLP systems

this is also part of evaluation

interpret the annotations!

carefully analyze whether it suits the purpose

when using a dataset, check its annotation documentation



Yes, crowdsourcing data collection and annotation are very convenient.

- redundancy & majority voting is possible
- can be less expensive and faster
- easier to get large ammounts of data

But that's not simple either.

Little to no control over participants, some of whom

- don't understand your task properly
- want to maximize money/hour earning
- are lazy
- claim to have a native language they do not

Participants are no experts and rely on instructions

- even though instructions might seem clear, it does not mean that random people understand it
- challenging to create intuitive instructions

Choosing crowdsourcing over expert labeling can mean trading quality for quantity

Need to tackle scamming and botting, e.g.:

- adding labeled samples to proof correctness
- adding fake-questions (where correct answer is known)

Need to validate quality, e.g.:

- redundancy: ask 2-5 different persons for the same label and compare

All that heavily increases cost per label! Take that into account when planning data collection.

- based on Fabian Galetzka's experience

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Evaluating annotation - a specific kind of evaluation

humans make mistakes, get distracted, forget the instructions… → many annotators annotate the same sample → they might disagree… → adjudication and metrics needed for evaluation

intra-annotator agreement: measures stability (consistency of each annotator alone over time)

inter-annotator agreement: measures reproducibility (different annotators on the same problem)

considered upper bound of human performance

Hovy & Lavid (2010)

Example metrics

- simple agreement
- Cohen's Kappa (corrected for chance)
- Fleiss' kappa (>2 annotators)
- Krippendorff's alpha

Illustration by undraw

See you next week!

- ☑ annotation in NLP

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